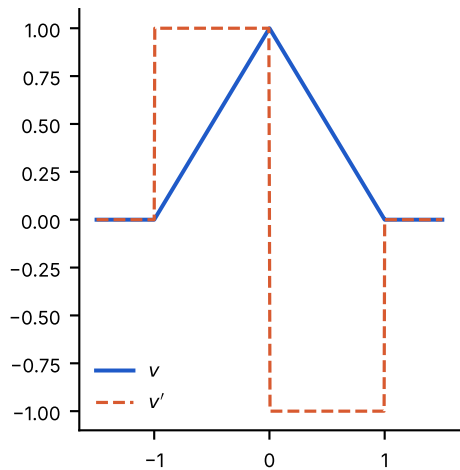
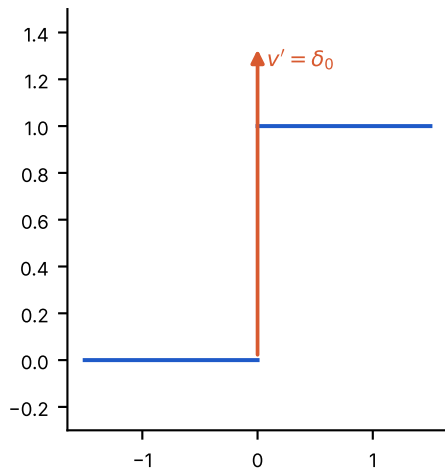
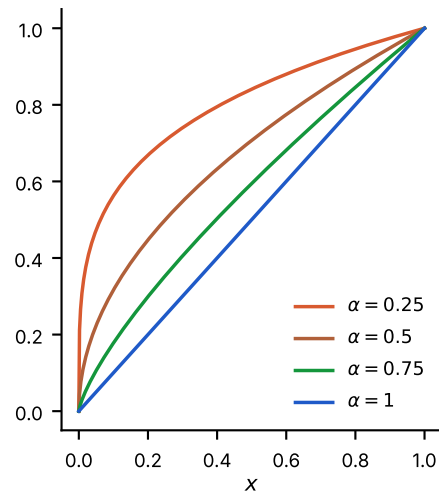
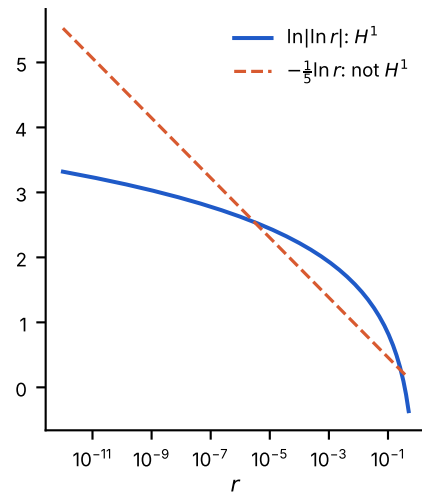
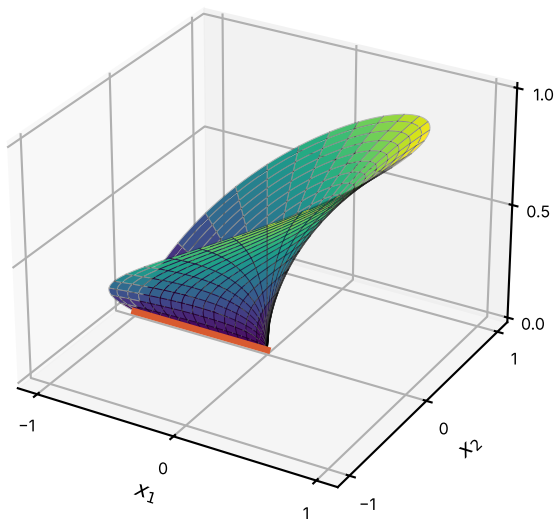


(a) hat:  $H^1$ , not  $H^2$ (b) step:  $L^2$ , not  $H^1$ (c)  $x^\alpha \in H^1$  iff  $\alpha > 1/2$ (d) unbounded, yet  $H^1$  (2D)(e) crack-tip field  $u = r^{1/2} \cos(\theta/2)$ ; crack in orange(f) level lines:  $|\nabla u| = \frac{1}{2\sqrt{r}}$ ,  $\int |\nabla u|^2 < \infty$ 